

-- step to configure nginx-reverse proxy server -
--

Important

- 1st == Check for Approval status
- 2nd == Before making any changes raise ticket to netbackup team for full " OS backup "
- 3rd == Take a snapshot of your of VM
- 4th == Make sure all the vm's Network Adapter are connected to Bridge [manual ip]

For nginx-reverse proxy server , we will build 3 servers i.e,

- 1= Dns server
- 2= Http server
- 3= nginx-reverse proxy server

For that we need 2 VM's

In VM 1 [Dns and http] server will be configure ..
And in Another Vm nginx-reverse proxy server ...

***** DNS-SERVER CONFIGURATION *****

-- steps to be follow --

1st SET HOSTNAME

```
# hostnamectl set-hostname dnssrv1.radical.com
```

```
[root@dnssrv1 ~]# hostnamectl set-hostname dnssrv1.radical.com
[root@dnssrv1 ~]# hostnamectl
Static hostname: dnssrv1.radical.com
    Icon name: computer-vm
    Chassis: vm 01F584
    Machine ID: 759a4db423d541ce87ccf6a098f90c89
    Boot ID: a4ded5a72e964ala92d3c5ccc041cc9d
    Virtualization: vmware
Operating System: Red Hat Enterprise Linux 9.2 (Plow)
```

2ND CONFIGURE NETWORK

```
# nmcli con mod ens160 ipv4.addresses 192.168.0.100/24

# nmcli con mod ens160 ipv4.method manual

# nmcli con up ens160

# ip a s
```

```
[root@dnssrv1 ~]# ip a s
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:1a:d6:73 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.0.100/24 brd 192.168.0.255 scope global noprefixroute ens160
        valid_lft forever preferred_lft forever
    inet 192.168.0.101/24 brd 192.168.0.255 scope global secondary noprefixroute ens160
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe1a:d673/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

4TH SET REPOSITORY [IF NOT THERE]

```
# yum repolist
```

```
[root@dnssrv1 ~]# yum repolist
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use s

repo id                                repo name
AppStream                              This is cdrom base repo
BaseOS                                 This is cdrom base repo
```

4TH MOUNT CDROM AND SEARCH FOR PACKAGES

```
# mount /dev/cdrom /mnt/

# yum search bind
```

```
[root@dnssrv1 ~]# yum search bind
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use subscription-manag

Last metadata expiration check: 4:31:55 ago on Mon 01 Jun 2026 06:24:45 AM EDT.
===== Name & Summary Matched: bind =====
bind.x86_64 : The Berkeley Internet Name Domain (BIND) DNS (Domain Name System) server
bind-chroot.x86_64 : A chroot runtime environment for the ISC BIND DNS server, named(8)
bind-dyndb-ldap.x86_64 : LDAP back-end plug-in for BIND
bind-libs.x86_64 : Libraries used by the BIND DNS packages
bind-license.noarch : License of the BIND DNS suite
```

5TH INSTALL THE PACKAGE

```
# yum install bind.x86_64 -y
```

6TH. SEARCH CONFIGURATION FILE

```
# rpm -qc bind
```

```
[root@dnssrv1 ~]# rpm -qc bind
/etc/logrotate.d/named
/etc/named.conf
/etc/named.rfc1912.zones
/etc/named.root.key
/etc/rndc.conf
/etc/rndc.key
/etc/sysconfig/named
/var/named/named.ca
/var/named/named.empty
/var/named/named.localhost
/var/named/named.loopback
```

7TH. BEFORE MAKING ANY CHANGES TAKE BACKUP

1st configuration file

```
# cp /etc/named.conf /tmp/named.conf_bkp1
# vi /etc/named.conf      [ put dns static ip ]
```

```
options {
    listen-on port 53 { 192.168.0.100; };
    listen-on-v6 port 53 { ::1; };
    directory      "/var/named";
    dump-file       "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named_stats.txt";
    memstatistics-file "/var/named/data/named_mem_stats.txt";
    secroots-file   "/var/named/data/named.secroots";
    recursing-file  "/var/named/data/named.recursing";
    allow-query     { any; };
}
```

8th Take backup of this file too before changes

2nd configuration file

```
# cp /etc/named.rfc1912.zones /tmp/named.rfc1912.zones_bkp1
```

```
# vi /etc/named.rfc1912.zones
```

“ add this line in the bottom ``

```
#####
zone "radical.com" IN {
    type master;
    file "fzone.radical.com";
    allow-update { none; };
};
```

9th Go to /var/named/ dir and copy named.localhost file to fzone.radical.com file

3rd configuration file

```
# cd /var/named/
```

```
# cp named.localhost fzone.radical.com
```

```
# vi fzone.radical.com
```

“ add this line ``

```
$TTL 1D
@      IN SOA  www.radical.com. root.www.radical.com. (
                                                0      ; serial
                                                1D      ; refresh
                                                1H      ; retry
                                                1W      ; expire
                                                3H      ; minimum
)

@      IN      NS       www.radical.com.
nginxsrv A      192.168.0.10
www      A      192.168.0.10
mail     A      192.168.0.10
~
```

[here we need to point Domain hostnames to the Proxy ip]

[The ip 192.168.0.10 is the static ip of nginx-reverse proxy server ip]

10th change group of fzone.radical.com file

```
# chgrp named fzone.radical.com
```

```
[root@dnssrv1 named]# ll
total 20
drwxrwx---. 2 named named 23 Feb 27 2023 data
drwxrwx---. 2 named named 60 Jun 1 10:30 dynamic
-rw-r-----. 1 root named 238 Jun 1 08:39 fzone.radical.com
-rw-r-----. 1 root named 2253 Feb 27 2023 named.ca
-rw-r-----. 1 root named 152 Feb 27 2023 named.empty
-rw-r-----. 1 root named 152 Feb 27 2023 named.localhost
-rw-r-----. 1 root named 168 Feb 27 2023 named.loopback
drwxrwx---. 2 named named 6 Feb 27 2023 slaves
```

11th Start the service

```
# systemctl enable --now named
```

```
[root@dnssrv1 ~]# systemctl status named
● named.service - Berkeley Internet Name Domain (DNS)
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-06-01 10:29:55 EDT; 2h 10min ago
     Process: 3007 ExecStartPre=/bin/bash -c if [ ! "$DISABLE_ZONE_CHECKING" == "yes" ];
     Process: 3009 ExecStart=/usr/sbin/named -u named -c ${NAMEDCONF} $OPTIONS (code=exi
   Main PID: 3010 (named)
      Tasks: 8 (limit: 10820)
     Memory: 29.2M
        CPU: 1.260s
```

12th stop firewall and SeLinux

```
# systemctl stop firewalld
```

```
# setenforce 0
```

***** HTTP-SERVER CONFIGURATION *****

Since we are on a same machine we will directly begin with the installation

1st Search for packages

```
# yum search httpd
```

```
[root@dnssrv1 ~]# yum search httpd
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use subscription-manager to register.

Last metadata expiration check: 0:12:21 ago on Mon 01 Jun 2026 11:01:37 AM
===== Name Exactly Matched: httpd =====
httpd.x86_64 : Apache HTTP Server
===== Name & Summary Matched: httpd =====
```

2nd Install the packages

```
# yum install httpd.x86_64 -y
```

3rd make two dir in /var/www/html/

```
# mkdir -p /var/www/html/radical
```

```
# mkdir -p /var/www/html/mail
```

```
[root@dnssrv1 ~]# cd /var/www/html/
[root@dnssrv1 html]# ls
index.html  mail  radical
[root@dnssrv1 html]#
```

4th Create index.html file for each

```
# vi /var/www/html/mail/index.html
```

“add this line”

```
***** welcome to mail.radical.com *****
```

```
~  
~
```

```
# vi /var/www/html/radical/index.html
```

```
"add this line``
```

```
***** WELCOME TO WWW.RADICAL.COM *****
```

```
~  
~
```

5th Make ip alias for the 2nd webside { mail.radical.com }

```
# nmcli con mod ens160 +ipv4.addresses  
192.168.0.101/24
```

```
# nmcli con up ens160
```

```
[root@dnssrv1 ~]# nmcli con mod ens160 +ipv4.addresses 192.168.0.101/24  
[root@dnssrv1 ~]# nmcli con up ens160  
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConne  
[root@dnssrv1 ~]# ip a s  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000  
    link/ether 00:0c:29:1a:d6:73 brd ff:ff:ff:ff:ff:ff  
    altname enp3s0  
    inet 192.168.0.100/24 brd 192.168.0.255 scope global noprefixroute ens160  
        valid_lft forever preferred_lft forever  
    inet 192.168.0.101/24 brd 192.168.0.255 scope global secondary noprefixroute ens160  
        valid_lft forever preferred_lft forever  
    inet6 fe80::20c:29ff:fela:d673/64 scope link noprefixroute
```

6th Create multiplewebside.conf file

```
# vi /etc/httpd/conf.d/multiplewebside.conf
```

```
"add this line``
```

```

<VirtualHost 192.168.0.100:80>
    ServerName radical.com
    ServerAlias www.radical.com
    DocumentRoot /var/www/html/radical/
    ErrorLog logs/www.radical.com-error_log
    CustomLog logs/www.radical.com-access_log common
</VirtualHost>

<VirtualHost 192.168.0.101:80>
    ServerName radical.com
    ServerAlias mail.radical.com
    DocumentRoot /var/www/html/mail/
    ErrorLog logs/mail.radical.com-error_log
    CustomLog logs/mail.radical.com-access_log common
</VirtualHost>

```

```
# httpd -t      [ check syntax ]
```

7th Start the service

```
# systemctl enable --now httpd
```

```
# systemctl status httpd
```

```

[root@dnssrv1 html]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-06-01 10:30:04 EDT; 52min ago
     Docs: man:httpd.service(8)
   Main PID: 3027 (httpd)
    Status: "Total requests: 4; Idle/Busy workers 100/0; Requests/sec: 0.00128; Bytes
      Tasks: 213 (limit: 10820)
     Memory: 19.2M
        CPU: 3.887s

```

8th Now do remote testing by running curl

```

[root@com1 ~]# curl 192.168.0.100
***** WELCOME TO WWW.RADICAL.COM *****

[root@com1 ~]# curl 192.168.0.101
***** welcome to mail.radical.com *****

```

***** NGINX-REVERSE PROXY SERVER *****

Go to another Vm [vm 2] for nginx-reverse proxy server configuration

===Before making any changes raise ticket to netbackup team for full “ OS backup”

=== Take a snapshot of your of VM

1st Set hostname

```
# hostnamectl set-hostname nginxsrv.radical.com
```

```
# hostnamectl
```

```
[root@nginxsrv ~]# hostnamectl
Static hostname: nginxsrv.radical.com
    Icon name: computer-vm
    Chassis: vm
    Machine ID: 1dc3aeb7cf4331983448248768241d
    Boot ID: a8a8b3ffd8aa4669b3ca62aa0f265c8c
    Virtualization: vmware
Operating System: Red Hat Enterprise Linux 9.2 (Plow)
    CPE OS Name: cpe:/o:redhat:enterprise_linux:9::baseos
    Kernel: Linux 5.14.0-284.11.1.el9_2.x86_64
    Architecture: x86-64
Hardware Vendor: VMware, Inc.
Hardware Model: VMware20,1
Firmware Version: VMW201.00V.24006586.B64.2406042154
```

2nd Configure Network

```
# nmcli con mod ens160 ipv4.addresses 192.168.0.10/24
```

```
# nmcli con mod ens160 ipv4.method manual
```

```
# nmcli connection up ens160
```

```
# ip a s
```

```
[root@nginxsrv ~]# ip a s
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default
    link/ether 00:0c:29:69:24:ba brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.0.10/24 brd 192.168.0.255 scope global noprefixroute ens160
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe69:24ba/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

3rd Set repository [If not there]

```
# yum repolist
```

```
[root@nginxsrv ~]# yum repolist
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use subscription-manager to register.

repo id                               repo name
AppStream                             This is file base repo
BaseOS                                This is file base repo
```

4th search for Packages

```
# yum search nginx
```

```
[root@nginxsrv ~]# yum search nginx
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use subscription-manager to

Last metadata expiration check: 4:41:02 ago on Mon 01 Jun 2026 06:55:24 AM EDT.
===== Name Exactly Matched: nginx =====
nginx.x86_64 : A high performance web server and reverse proxy server
===== Name & Summary Matched: nginx =====
nginx-all-modules.noarch : A meta package that installs all available Nginx modules
nginx-core.x86_64 : nginx minimal core
```

5th Install the Packages

```
# yum install nginx.x86_64 -y
```

6th Create a file reverse-proxy.conf

```
# vi /etc/nginx/conf.d/reverse-proxy.conf
```

```
server {
    listen 80;
    server_name www.radical.com;

    location / {
        proxy_pass http://192.168.0.100:80;
    }
}

server {
    listen 80;
    server_name mail.radical.com;

    location / {
        proxy_pass http://192.168.0.101:80;
    }
}
```

```
# nginx -t [ check syntax ]
```

7th Start the service

```
# systemctl enable --now nginx
```

```
# systemctl status nginx.service
```

```
[root@nginxsrv ~]# systemctl status nginx.service
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-06-01 10:30:21 EDT; 3h 17min ago
     Process: 2700 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
     Process: 2701 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 2703 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
   Main PID: 2704 (nginx)
      Tasks: 3 (limit: 10820)
     Memory: 3.9M
        CPU: 69ms
   CGroup: /system.slice/nginx.service
           └─2704 "nginx: master process /usr/sbin/nginx"
             └─2708 "nginx: worker process"
               └─2709 "nginx: worker process"
```

8th Stop firewall and SeLinux

```
# systemctl stop firewalld.service

# setenforce 0
```

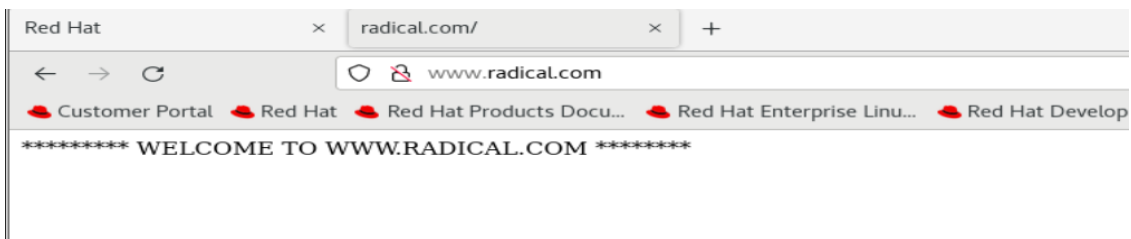
Now, do remote testing from client machine

1st Client side make an entry in this file /etc/resolv.conf

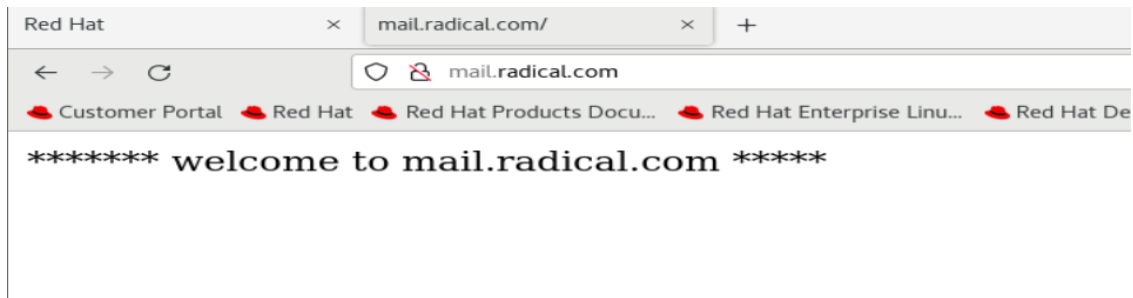
```
# vi /etc/resolv.conf
```

```
# Generated by NetworkManager
nameserver 192.168.0.100
~
..
```

2nd Open firefox and check



mail.radical.com



[Verify nginx-reverse proxy server is working or not]

[Here you need to see the nginx-reverse proxy server ip not the web-server ip]

dig command output for www.radical.com web-server

```
[root@com1 ~]# dig www.radical.com
; <<>> DiG 9.11.36-RedHat-9.11.36-8.el8 <<>> www.radical.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 2720
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 793f49307456633a010000006alda84796196b6be7a80282 (good)
;; QUESTION SECTION:
;www.radical.com.                IN      A
;; ANSWER SECTION:
www.radical.com.                86400   IN      A      192.168.0.10
;; Query time: 11 msec
;; SERVER: 192.168.0.100#53(192.168.0.100)
;; WHEN: Mon Jun 01 21:12:00 IST 2026
;; MSG SIZE  rcvd: 88
```

dig command output for mail.radical.com web-server

```
[root@com1 ~]# dig mail.radical.com
; <<>> DiG 9.11.36-RedHat-9.11.36-8.el8 <<>> mail.radical.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 60893
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: c96d4fe9eeb57758010000006alda887065035585fb3945f (good)
;; QUESTION SECTION:
;mail.radical.com.              IN      A
;; ANSWER SECTION:
mail.radical.com.              86400   IN      A      192.168.0.10
;; Query time: 5 msec
;; SERVER: 192.168.0.100#53(192.168.0.100)
;; WHEN: Mon Jun 01 21:13:03 IST 2026
;; MSG SIZE  rcvd: 89
```

nginx-reverse proxy server is working find ..

```

[root@com1 ~]# curl -I http://www.radical.com
HTTP/1.1 200 OK
Server: nginx/1.20.1
Date: Mon, 01 Jun 2026 14:36:16 GMT
Content-Type: text/html; charset=UTF-8
Content-Length: 46
Connection: keep-alive
Last-Modified: Mon, 01 Jun 2026 10:22:50 GMT
ETag: "2e-6532e939d2518"
Accept-Ranges: bytes

[root@com1 ~]#
[root@com1 ~]# curl -I http://mail.radical.com
HTTP/1.1 200 OK
Server: nginx/1.20.1
Date: Mon, 01 Jun 2026 14:36:54 GMT
Content-Type: text/html; charset=UTF-8
Content-Length: 37
Connection: keep-alive
Last-Modified: Mon, 01 Jun 2026 12:25:47 GMT
ETag: "25-653304b54fe18"
Accept-Ranges: bytes

```

Now , set the firewall rules and enforce the Selinux in both the vm's

[Vm 1] dns and http server side

```

# systemctl restart firewalld.service
# firewall-cmd --permanent --add-service=dns
# firewall-cmd --permanent --add-service=http
# firewall-cmd -reload
# firewall-cmd --list-all

```

```

[root@dnssrv1 ~]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: ens160
  sources:
  services: cockpit dhcpv6-client dns http ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:

```

SeLinux

```
[root@dnssrv1 ~]# setenforce 1
[root@dnssrv1 ~]# getenforce
Enforcing
```

[vm 2] nginx-reverse proxy server side

```
# firewall-cmd --permanent --add-service=http
# firewall-cmd -reload
# firewall-cmd --list-all
```

```
[root@nginxsrv ~]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: ens160
  sources:
  services: cockpit dhcpv6-client http ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
```

SeLinux

```
[root@nginxsrv ~]# setenforce 1
[root@nginxsrv ~]# getenforce
Enforcing
```

Now, do Final remote testing after applying firewall rules and enforcing SeLinux.

Conclusion:

Successfully configured DNS, NGINX Reverse Proxy, and Apache Web Server.
Verified domain resolution and web access through the reverse proxy setup.